

INSTRUCTIONS SET

S.NO	INSTRUCTIONS	OPCODE	HEXCODE
1.	Addition of two registers	ADD	6'b000000
2.	Addition with immediate data	ADDI	6'b000001
3.	Subtraction of two registers	SUB	6'b000010
4.	Subtractions with immediate data	SUBI	6'b000011
5.	Bitwise logical AND operation	AND	6'b000100
6.	Bitwise logical OR operation	OR	6'b000101
7.	Bitwise logical XOR operation	XOR	6'b000110
8.	Bitwise logical AND operation with immediate data	ANDI	6'b000111
9.	Bitwise logical OR operation with immediate data	ORI	6'b001000
10.	Set if less than (on comparing with registers)	SLT	6'b001001
11.	Set if less than (on comparing with immediate data)	SLTI	6'b001010
12.	Multiplications two registers	MUL	6'b001011
13.	Shift left by one bit	SLL	6'b001100
14.	Shift left by immediate data	SLLI	6'b001101
15.	Shift right by one bit	SRL	6'b001110
16.	Shift right by immediate data	SRLI	6'b001111
17.	Load the immediate data into register	LW	6'b010000
18.	Store the register into immediate data	SW	6'b010001
19.	Set if equal (on comparing with registers)	SEQ	6'b010010
20.	Set if not equal (on comparing with registers)	SNEQ	6'b010011
21.	Taken Branch if not equal to zero(register)	BNEQZ	6'b010100
22.	Taken Branch if equal to zero(register)	BEQZ	6'b010101
23.	Taken Branch if two registers are equal	BEQ	6'b010110
24.	Taken Branch if two registers are not equal	BNEQ	6'b010111
25.	Bitwise logical XOR operation with immediate data	XORI	6'b011000
26.	Jump to immediate address	JMP	6'b011001
27.	Halt :end the program	HLT	6'b111111